

Adding value with CLEWS – Modelling the energy system and its interdependencies for Mauritius



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H I G H L I G H T S

- Climate, Land-use, Energy and Water Systems (CLEWS) are strongly interlinked.
- Related policy making is often informed by disconnected resource assessments.
- The value of interlinking such assessments is demonstrated, focusing on energy.
- The effects of water stress on the energy system were modelled more accurately.
- The benefit of ethanol generation was largely overestimated without interlinkages.

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Climate, Energy, Water and Land-use Systems (CLEWS) are closely integrated. Yet, most related decision and policy making occurs in disparate institutional entities, informed by relatively disconnected assessments of the individual resource systems. This paper presents the added value of an integrated analytical assessment approach. In doing so, it explicitly values various interdependencies and interactions between CLEWS primarily from an energy sector perspective. The island state of Mauritius was identified as a useful case study given its diverse climate, its increasing water stresses, and its policy focus on reshaping agricultural land-use and reducing fossil fuel imports. Several scenarios to 2030 were defined and analysed to demonstrate the tensions around the CLEWS nexus. Results from an assessment of the energy system with no modelled interlinkages to land-use, energy and water systems are first presented. Then, these are compared to those from an integrated CLEWS assessment. This serves to highlight important dynamics that would have been overlooked without such a systems approach. As an example, the added value of this approach is clearly demonstrated when rainfall reductions are taken into account, and where future land-use changes might occur.

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1. Introduction

1.1. Rationale

Globally, the most basic human needs rely on the availability of a few key resources. Those include: water for drinking; water and land for food production; and energy for services like lighting, cook-

ing and heating. All of these resources have in common that they are – depending on the context – constrained in some way. As demand for these resources tends to increase globally, so do efforts to access and secure them, often leading to tensions in areas where they are scarce [1]. At a national level, such tensions may be due to a dependency on resource imports. At a more local level, they may be due to diverging priorities on how they should be used. The prospects of climate change may exacerbate related stresses. This is due to the potential changes in rainfall and in the availability of arable land. These changes may be accompanied by increases in energy

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demand, e.g., for cooling or desalinating water. Examples for stresses which are related to resources include tensions regarding the use of the river Nile [2,3], deforestation of the Amazon rain forest and its implications for climate change [4], or gas exports from Russia to Europe via the Ukraine in 2009 [5,6].

Any single one of these resources might cause stresses to secure their accessibility. Even more complex situations may evolve in cases where Climate, Energy, Water and Land-use Systems (CLEWS) are interrelated. For example, future rainfall reductions in Mauritius would affect the water availability on this water stressed island, e.g., for hydropower generation and agriculture. Maintaining agricultural production requires an increased electricity generation for groundwater pumping for irrigation. Meeting this increased electricity demand and compensating the lower hydropower production requires additional generation by other power plants. Increasing the generation of fossil fuel-fired power plants increases the country's import dependency and greenhouse gas emissions. Further examples of the relevance of capturing such interlinkages between resource systems are provided in [1].

Government structures and the associated division of responsibilities and priorities do not favour the integrated approach required to capture these interlinkages. Historically, related decision-making is often based on fragmented assessments of CLEWS resources and interactions between all resource systems are rarely taken into account [7]. From an energy modelling perspective, related assessments hardly ever go beyond a consideration of greenhouse gas emissions or biomass as a fuel for energy generation (e.g., as in [8–14]). [15] goes one step further by interlinking a water demand model with models representing a PV pumping system and in [16] elements of a water system model were incorporated into an energy model.

A more holistic assessment was provided in [17], which focusses on the intensification of agriculture in Burkina Faso to help ensure future food production. The study indicated the importance of an integrated CLEWS approach when designing strategies in support of sustainable development.

1.2. About this paper

This paper quantitatively demonstrates the added value of such an integrated CLEWS assessment. This is done by comparing conclusions derived from an energy model with those of an integrated CLEWS approach. Mauritius was identified as an ideal case study given its diverse climate, its growing water stress, and its focus on reshaping agricultural land-use and decreasing fossil fuel imports. Further, Mauritius has a robust data set on the use of its resources¹. As a small island, it also has conveniently defined system boundaries. The choice of Mauritius allowed building on a previous case study, which identified important CLEWS dynamics for Mauritius [19]. In line with the long-term energy strategy of Mauritius [20], increases in local ethanol production are investigated in and serve to demonstrate the CLEWS nexus.

While this paper considers the relevance of an integrated CLEWS assessment, it does so through an “energy perspective”. It feeds into a related research effort with a broader focus, which provides more detail on the overall CLEWS approach and its application on Mauritius [21]. This related effort presents further background on the applied climate, land-use and water models. It maps the interactions between the individual resources models and shows overall results including greenhouse gas emissions, water supply dynamics and crop productivities. For a more conceptual background on the CLEWS approach, please refer to [7].

This paper first provides a brief background on Mauritius in Section 2 before presenting the methodology and scenarios in Sections

3 and 4. Results for these scenarios are presented in Section 5. The added value of the CLEWS approach is synthesized at the end of Section 5. The paper concludes in Section 6 by highlighting conditions under which the benefits of an integrated CLEWS approach are likely to justify the additional effort required.

2. A brief background on Mauritius

2.1. Mauritius and its economy

The Republic of Mauritius is an archipelago in the Indian Ocean, 950 km east of Madagascar². Its population amounts to 1.2 million inhabitants. With a population density of 668 people per km², it is the most densely populated country in Africa [22–24].

Mauritius is classified as an upper-middle-income country by the World Bank. Its economy is primarily based on services, which contribute 69% to the gross domestic product (GDP). 27% of GDP are generated in industry including sugar processing. About 90% of the sugar produced is exported to the European Union (EU). Sugar represented 11% of total domestic exports in 2009 [25–28].

The country's economic and social progress is potentially under threat from external shocks. In 2009, an EU decision came into effect which cut the guaranteed sugar import price by 36% compared to 2006 price levels. Further, Mauritius imports coal and liquid fuels to meet 83% of its energy needs. It is therefore very vulnerable to rising and volatile world energy prices. Increased energy security and diversifying income from exports are key policy concerns [20,29–32].

In the wake of the reform of the EU sugar import regime³, the government has formulated several measures to refocus agriculture. This is reflected by a shift from the farming of sugar cane to food crops. Further, since 2004, several sugar cane mills have started to produce ethanol. The target production of 30 million liters from molasses, either for domestic blending with gasoline or export, is therefore well in reach [23,34–39].

2.2. Mauritius and its CLEW resources

2.2.1. Some background on climate, agricultural land-use and water

Mauritius has a sub-tropical maritime climate with an average annual temperature between 22 and 33 °C. Although it is relatively small with an area of 186,500 ha, its rainfall patterns are diverse and characterized by its topography. They are strongly dependent on elevation, proximity to the coast, and position relative to the prevailing winds and mountain ranges. Yearly averages may vary from as low as 750 mm in the western areas up to over 4000 mm on the central plateau. Summer, lasting from November to April, receives two-thirds of the yearly rain [22,40,41].

These rainfall patterns are reflected in diverse agricultural conditions and result in extensive water transportation needs for irrigation. Full control irrigation increased from 12,000 ha in 1970 to 19,900 ha in 2010. More of 90% of this irrigated area is used for sugar cane production. Sugar cane is currently processed by six sugar producing factories, each with its dedicated cropland. Overall, 34% of the total area of Mauritius is cultivated [23,42].

Water demand per capita amounts to 221 l/capita/day. 63% of all water consumption can be attributed to agriculture⁴. While the agricultural sector currently meets its water demand mainly

¹ E.g., via the website of the Central Statistics Office [18].

² All data mentioned in this paper refers to the main island of Mauritius. Small surrounding islets as well as the neighbouring island of Rodrigues have not been included in the analysis. Their total area constitutes less than 10% of the total area of Mauritius (in total: 2040 km²) [22].

³ The EU decision to reduce its guaranteed sugar import prices was taken to comply with its commitments within the World Trade Organization (WTO). Mauritius was considered as one of the countries suffering most from this reform [33].

⁴ Utilisation of water for hydropower is not taken into account.

with surface water, over half of all other water demand is met through extraction from boreholes⁵. Increasing this ground water use would be problematic: abstractions have reached saturation levels in large parts of the country with aquifers being at risk of sea water intrusion. This can especially be an issue in the northern and eastern coastal areas and during dry spells. While Mauritius is increasingly getting water stressed, water conservation measures, wastewater treatment and desalination are becoming popular. For example, hotels are obliged to provide related provisions since 2005 [23,30,43–45].

2.2.2. Mauritius' energy system

Mauritius imports all of its petroleum products as well as coal. It uses only limited amounts of domestic renewable energy resources. These include fuel wood, wind power, hydropower and biomass – especially bagasse, a by-product of sugarcane processing. Bagasse accounts for 93% of the energy content of all domestic energy resources and is used for co-generation of heat and electricity at sugar factories [30,46,47].

Over the past decade, Mauritius' dependence on energy imports has grown from approximately 1100 ktoe in 2000 to about 1500 ktoe in 2010. The corresponding share of energy imports has increased from 11% of the total import bill in 2000 to 18% in 2010. During the period of peak oil prices in 2008 it rose to even 21% [30,41,48–50].

Electricity generation utilizes 96% of all coal imports. The remainder is used by the manufacturing industries. Gasoline, diesel and aviation fuel are the three main transportation fuels. Overall, the transport sector accounts for 54% of all demand for petroleum products, while 26% are used for electricity generation [30].

In 2010, 2689 GWh of electricity were generated. Thermal power plants, including the incineration of bagasse, contributed with 96% to the total electricity generation. Hydro and wind power provided to the remaining 4%. The peak demand in 2010 was 404.1 MW. The generation fuel mix has been evolving over time with a major shift from fuel oil to coal. Of the 778 ktoe of fuel inputs used for power generation in 2010, coal comprised 51%, oil products 25% and bagasse 23% [30,48].

2.2.3. 'Medine' and 'F.U.E.L.'

Special attention was paid to the sugar cane processing plants 'Medine' and 'F.U.E.L.', where ethanol production was introduced in some scenarios. Medine is situated in the more water stressed western part of Mauritius, some 15 km south-west of the capital Port Louis. In 2010, it processed the sugar cane of 4600 ha of land. This corresponds to 8% of the total area in Mauritius where sugar cane is harvested. The by-product bagasse is used to generate heat and electricity for its own use and for export to the national grid. Its current electricity generation capacity is 6 MW [42,51].

F.U.E.L. is located about 20 km to the east of Port Louis. In 2010, it processed the sugar cane of 13,800 ha, corresponding to 24% of the total area harvested. It produces electricity from both, bagasse and coal, at a maximum capacity of 27 MW. Like at Medine, waste heat from bagasse is used for sugar cane processing. With 1.4 million tonnes, the sugar cane production of both plants amounts to 30% of the total production of Mauritius. The combined electricity generation of 167 GWh provides 7% to the total generation [42,51].

3. Methodology

Most decision and policy making related to land-use, energy and water systems occur in disconnected institutional entities with little, if at all, coordination or communication [7]. Therefore several

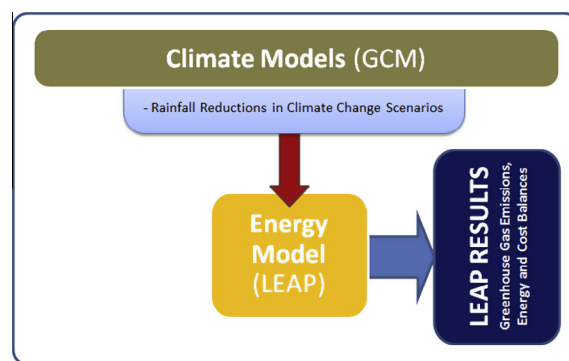


Fig. 1. Schematic of Current Practice approach.

scenarios were first analysed with a focus on the energy system and without taking explicit interlinkages between land-use, energy and water systems into account. Rainfall reductions based on publicly available climate model data were considered as an external input value to estimate hydropower availability for scenarios considering the effects of climate change [52,53]. This initial set up is outlined in Fig. 1 and referred to as the 'Current Practice' approach.

Next, the same scenarios were re-assessed considering these interlinkages by using the integrated 'CLEWS approach' as shown in Fig. 2. This approach draws on the following individual well-tested and specialised resource models.

3.1. Modelling tools

3.1.1. Climate

External climate models were not set up as an integral part of this assessment. Instead, selected General Circulation Models (GCM)⁶ and their corresponding projections were used to derive temperature and rainfall assumptions, which were applied to the other resource models [52,53]. Greenhouse gas emissions were accounted for in the energy model, but not fed back into the climate models as they on their own were not assumed to affect the local climate conditions.

3.1.2. Land-use

The modelling of the land-use system draws on the raster-based Agro-Ecological Zones land production planning model (AEZ) [53]. A resolution of approximately 250 m time 250 m was used to derive the production potential of the farmland used for ethanol production. Further, AEZ served to calculate irrigation requirements under different climate conditions as well fertilizer input required by different crops under different conditions like crop cycles per year.

3.1.3. Water

The water system was modelled using the Water Evaluation and Planning System (WEAP) [54] tool. WEAP is a tool for water resource planning, which was applied to assess the implications of local municipal and agricultural water requirements on the national water supply schemes. Within WEAP, Mauritius' rivers were modelled as about 60 catchment areas. Each of them was characterized by specific hydrological and climatic profiles and land-cover classes, interlinked with its reservoirs and the five main aquifers of the island.

Further information about the interlinkages between WEAP and AEZ as well as background on the structure and set up of these models are presented in [21].

⁶ Including the following models: C2A2, CSA2, EHA2, C2B2, CSB2, EHB2, H3B2, CSB1, H3B1 and H3A1.

⁵ Again, hydropower is excluded.

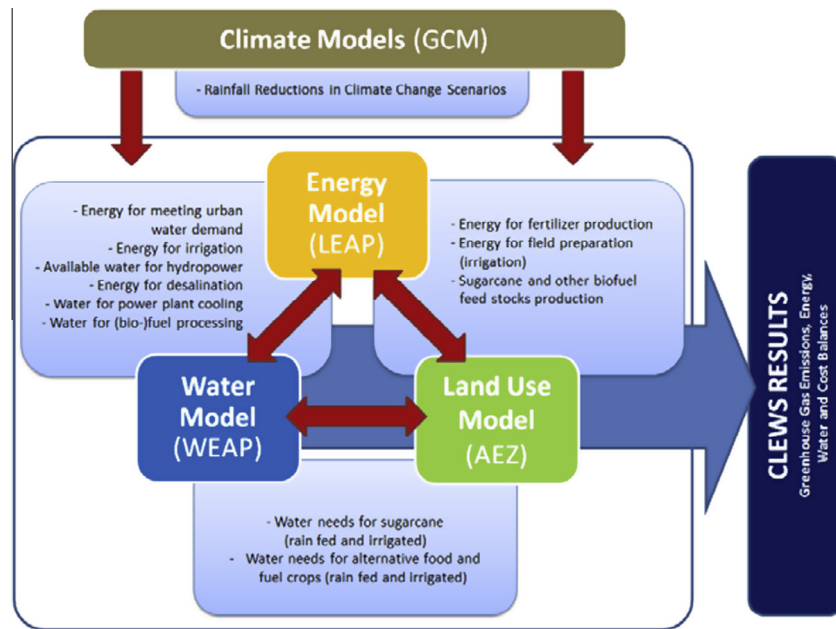


Fig. 2. Schematic of CLEWS approach.

3.1.4. Energy

The energy system was assessed with the Long-range Energy Alternatives Planning (LEAP) tool. LEAP is a widely used tool for analysing energy systems, including power dispatch and capacity expansion. It enables a consideration of various economic sectors, technologies, costs and emission profiles [55–57]. The LEAP tool was set up to reflect the extraction, conversion and demand of energy.

3.2. 'Current Practice approach'

In the Current Practice approach, the LEAP tool was used to calculate:

- the average⁷ power plant dispatching and future capacity requirements;
- the ethanol production from sugar cane based on data regarding historic sugar cane harvests;
- changes in fuel imports to the island due to the substitution of gasoline with ethanol;
- the effects of changes in rainfall patterns on generation; as well as
- greenhouse gas emissions, both on the island, as well as associated external emissions due to fuel processing and fertiliser supply to the island⁸.

The energy system was set up based on historical demand data from the period 2005–2009 and generation data from 2005–2008 [48,50,59–64]. Aligned with historical growth rates, an electricity

demand growth of 3.5% was assumed. Any additional electricity requirements for pumping for irrigation and desalination were explicitly entered in the model. Economic considerations are based on an assumed oil price of 80 USD per barrel, a coal price of 60 USD per ton⁹ and a sugar export price of 420 USD per ton¹⁰.

To model the electricity generation, all power plants and co-generating processing plants that export electricity to the national grid were modelled individually. Future power expansions plans were taken into account [20,65]. Any additional future capacity needs were assumed to be met by investments in coal-fired power plants to meet the base load and oil-fired power plants to meet balancing requirements¹¹. Efficiencies and capacity factors of power plants were calibrated from historical data. Power plants were dispatched giving priority to those with the lowest short run marginal generating costs. Capital, operating and fuel costs were chosen according to data based on assessments of comparable international plants [66,67]. Please refer to the annex for an overview of key power plant input data.

While hydropower generates less than 5% of the total electricity, it is strongly affected by the rainfall reductions in scenarios considering climate change. This is due to reduced inflows, potentially increased reservoir outflows and diversions to meet other water demands in times of shortage. A hydropower plant connected to a reservoir is assumed to be able to generate electricity in all months where the storage volume is more than an assumed 'dead storage' capacity of 5% of the total. A yearly 'hydro factor' was calculated. This factor de-rates the hydropower generation should

⁷ The temporal information within LEAP is based on time slices. These time slices represent times of the year with specific demand or generation characteristics. The accuracy of the dispatching modelled with LEAP therefore relates to the chosen number and definition of time slices as opposed to the detailed temporal resolution required for the daily dispatching by a system operator. Refer to [58] for a view on the value of an increased temporal resolution.

⁸ Including those associated with oil refining, coal processing and fertiliser production. Note that in scenarios where sugar cane is used for ethanol instead of sugar production, external economic effects outside of Mauritius were not considered. Those could be significant, yet are difficult to assess. For example, the loss of area for sugar cane farming could be compensated by increases in farm land in other sugar producing countries. This could potentially lead to deforestation and associated greenhouse gas emissions.

⁹ Whenever ton is used as a unit in this paper, it refers to metric tons.

¹⁰ At these prices the losses associated with reduced sugar exports are lower than the gains from the reduced imports of gasoline. Prices vary and are likely to be volatile in global markets. The positive cost-benefit balance is therefore not guaranteed. For an oil price of 100 USD/barrel and sugar prices higher than 700 USD/ton, ethanol production would not be profitable any longer [7]. This paper focuses on a comparison of differences in key energy dynamics with and without an integrated CLEWS approach. A detailed sensitivity analysis of the costs is outside of its scope.

¹¹ Coal-fired power plants were assumed to contribute 70% of future capacity requirements for meeting peak demand. Oil-fired power plants were assumed to provide the remaining 30%, plus requirements to meet the system's reserve margin of 21%. For balancing requirements, the minimum share of oil based generation in the total mix was set to 15%. This compares to a national target to reduce the share of oil to 20% in 2025 [20].

the times with low storage volumes increase in the future. This could be due to the future impact of climate change or increases in reservoir outflows. Smaller hydropower plants which are not connected to reservoirs are assumed to reduce their generation by the same share that the average river flow is reduced. The monthly storage volumes and river flows were calculated by assuming river flow reductions to equal the expected rainfall reductions in per cent. No increases in competing water uses were considered in the 'Current Practice approach', e.g., for agricultural, municipal or industrial water demands.

3.3. 'CLEWS approach'

When reassessing the energy system considering the CLEWS approach, additional interlinkages between the energy model and the land-use and water model were taken into account¹². The required steps for this assessment were as follows:

1. Identify the interactions between the climate, land-use, energy and water resource systems. This first step can be considered as the most important step, and might require some form of collaboration with experts on the various resource systems.
2. Quantify these interactions. Such quantification focused, for example, on the groundwater demand for irrigation in agriculture and the associated energy requirements for pumping. Quantifying the interactions is required to get a sense of the relevance of the various interactions.
3. Represent the interactions within the modelling framework.
4. Clearly define the required exchange of data between the resource models and its format. This is referred to as "soft-linking" the individual models. It is only required in case separate modelling tools are applied.
5. Calibrate the modelling tool(s) based on historical data. Conveniently, this is first done through separated model runs, without considering any interlinkages between the models.
6. Develop scenarios to compare the implications of various key assumptions on future development pathways.
7. Represent the scenarios in the modelling tool(s).
8. As some outputs from one model might serve as input data for another, iterations may be required before a convergent solution emerges.
9. Interpret results and the differences between the Current Practice and the CLEWS approach.

Fig. 2 provides an overview of the interactions between these tools.

While the structure of the LEAP model was kept exactly the same as in the Current Practice approach, the following interlinkages to the other resource models were taken into account:

- The bioenergy production potential and fertiliser requirements for growing an alternative crop were derived from the land-use model.
- Additional¹³ water pumping demands were incorporated as input values to the energy model. Those demands include pump-

Table 1

Overview of abbreviations for all scenarios.

	Business as usual	Ethanol –1 st generation	Ethanol –2nd generation
Current Practice approach	BAU	1GEN	2GEN
CLEWS Approach	BAU + CC	1GEN + CC	–
	BAU ^{CLEWS}	1GEN ^{CLEWS}	2GEN ^{CLEWS}
	BAU + CC ^{CLEWS}	1GEN + CC ^{CLEWS}	2NC + CC ^{CLEWS}

ing for urban water supply and sugarcane processing as derived from the water model, and pumping for agricultural irrigation as derived via the land-use model.

- Desalination demand for urban water supply was included as another input value to the energy model.
- Monthly storage volumes and river flows were derived from the water model. This enabled a consideration of competing water demands like agricultural or municipal demands. The hydropower generation was calculated based on these storage volumes and river flows following the same logic as in the Current Practice approach.
- Water demands for ethanol production and power plant cooling requirements were derived as output data from the energy model and fed into the water model.

4. Scenarios

In line with the long-term energy strategy of Mauritius, the effects of increases in local ethanol production were investigated up until 2030. Ethanol was assumed to be added to the fuel mix of the local car fleet, or exported if deemed beneficial¹⁴. Changing weather patterns and decreasing rainfall are of concern to the Government of Mauritius as the island is prone to drought [69]. Therefore, the effects of climate change were considered.

In addition to a "business as usual" case based on historical trends and current government targets for renewable electricity generation, the scenarios to model these developments were grouped as follows:

- Scenarios without Climate Change Considerations: The sugar cane processing plants 'Medine' and 'F.U.E.L.' are converted to produce ethanol instead of sugar from 2015 onwards.
- Scenarios with Climate Change Considerations: Additionally, the effects of climate change are simulated by decreasing rainfall linearly to 20.4% during the period from 2010 to 2030. This is based on a worst case scenario derived from the selected General Circulation Models (GCM) [52,53].

The following section outlines the scenarios established based on these considerations. If possible, the exact same scenarios were set up for both, the Current Practice (Section 4.1) and the CLEWS assessments (Section 4.2). Only one scenario explicitly relies on an integration of resource systems. It was therefore only possible to assess this scenario pursuing the CLEWS approach.

Table 1 provides an overview of all of the scenarios assessed and their abbreviations. Note that the superscript 'CLEWS' refers to the assessment of a scenario based on the CLEWS approach. Scenarios without this superscript were assessed based on the Current Practice approach. The extension '+CC' refers to scenarios considering climate change.

¹⁴ This may occur if the car fleet is not able to absorb all of the ethanol produced. In this case, it was assumed that there is an external market for ethanol from Mauritius and that import prices of gasoline per energy content equal export prices for ethanol. Therefore, it has no economic implications if the ethanol is consumed on Mauritius or exported. [68] reports that such exports are already taking place, with an ethanol producer aiming at increasing its exports to 30 million litres per year.

¹² While an energy model on its own can be developed relatively straightforward based on data published in literature, pursuing the CLEWS approach comes with a considerable level of effort. When drawing on individual resource models, the output data of one model might serve as input data for another. Agreeing on a clear timeline, resolution and format of these inputs and output is therefore a necessity for a successful CLEWS assessment. Further, common assumptions and scenarios have to be well thought through and agreed on upfront. Any later change is a potential cause of errors if not implemented correctly in all models and scenarios, and might cause significant delays if not scheduled for.

¹³ Additional to the demand in the business as usual scenario of the Current Practice approach (BAU).

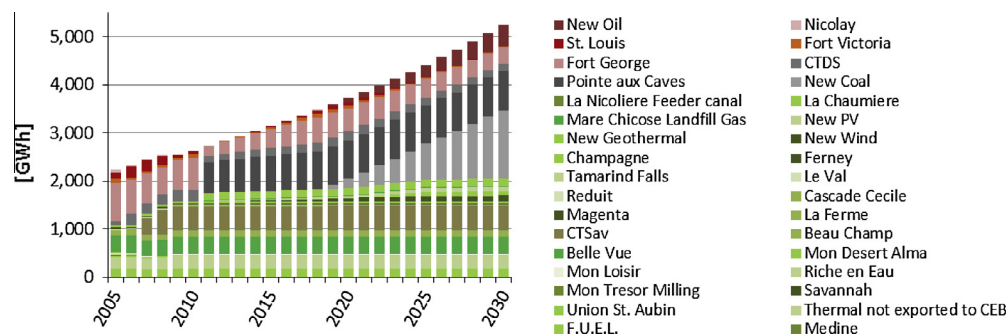


Fig. 3. Generation Mix by Power Plant for the business as usual (BAU). Red areas are assigned to power plants which are fired with petroleum products, grey areas to coal-fired power plants and green areas to renewable energy, including plants running on both bagasse and coal. This colour coding is maintained throughout this paper. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.1. Current Practice approach

4.1.1. Scenarios without climate change considerations

4.1.1.1. BAU: business as usual. In this scenario, electricity and gasoline demand growth follow historical trends. Electricity and heat generation from bagasse continues at current levels, but no ethanol is produced. Future renewable electricity generation shares reflect the targets outlined in the long-term energy strategy of Mauritius. Rainfall patterns, reservoir levels and thus hydropower availability were expected to reflect historical levels. The results of the following scenarios assessed with the Current Practice approach are illustrated as changes to this baseline scenario.

4.1.1.2. 1GEN: ethanol – 1st generation. Sugar production is changed to so-called ‘first generation’ ethanol production¹⁵. Ethanol is blended with gasoline to meet domestic transportation fuel demand or used for export. The by-product bagasse is used to produce electricity at the sugar cane processing plants. Excess electricity is sold to the national grid.

4.1.1.3. 2GEN: ethanol – 2nd generation. Sugar cane is again used to produce ethanol. However, excess bagasse is no longer used to produce electricity for the national grid. Instead, it is also converted to ethanol via hydrolysis, using so called ‘second generation’ technologies¹⁶.

4.1.2. Scenarios with climate change considerations

4.1.2.1. BAU+CC: business as usual, water stress. This scenario builds on the business as usual (BAU) scenario, but with rainfall reductions due to climate change. No ethanol is produced.

4.1.2.2. 1GEN+CC: ethanol – 1st generation, water stress. This scenario builds on the first generation ethanol production scenario (1GEN), but with rainfall reductions due to climate change¹⁷.

4.2. CLEWS approach

All scenarios assessed with the Current Practice approach are reassessed with the CLEWS approach, i.e., taking the interlinkages with the water and land-use models into account. The results are presented as changes to the respective baseline scenario

(BAU^{CLEWS}). Further, an additional scenario is investigated, which could not have been assessed pursuing the Current Practice approach:

4.2.1. 2NC+CC^{CLEWS}: ethanol – 2nd generation, new crop, water stress, CLEWS approach

This scenario builds on the second generation ethanol production scenario (2GEN^{CLEWS}). It explores the potential of growing an alternative crop as ethanol feedstock for processing at the two selected processing plants. The characteristics of this alternative crop were aligned with those of corn. It allows for two harvest cycles and is more drought resistant than sugar cane. All other processing plants continue to rely on sugar cane for sugar production. As for all climate change scenarios, rainfall is reduced based on the outlined climate change assumptions.

4.3. Assumptions related to ethanol production, agriculture and water supply

If not stated otherwise, sugar cane production is maintained at historical levels in all scenarios¹⁸. When applying the CLEWS approach, the water supply and land-use models were set up accordingly to support this level of production: When precipitation is insufficient, irrigation is used to meet any additional water requirements for feed stock. The irrigation water demand is covered by surface runoff. Should this not suffice, groundwater pumping is considered. Urban water demand is met through rivers and reservoirs, groundwater pumping or with [supplementary desalination](#)¹⁹. Desalination is considered for meeting urban demand if pumping would reduce the available groundwater by more than 15% below the historical minimum. Groundwater is already currently at risk of saline intrusions. Therefore, this appears as a rather conservative assumption with regard to demand increases for desalination.

5. Results

Based on the outlined methodology, this section presents the results for the business as usual, the first and the second generation scenarios. Each of them is first assessed applying the Current Practice approach with and without climate change considerations, followed by a reassessment pursuing the CLEWS approach. Changes in the results between different scenarios and the business as usual case are presented for the year 2030. This serves to assess what level of detail is gained and to what extent the derived

¹⁵ ‘First generation’ refers to ethanol production from sugar and starch crops [70].

¹⁶ ‘Second generation’ refers to ethanol production from lignocellulosic biomass. This extends the potential feed stock sources, e.g., to bagasse from sugar cane, waste products from agriculture and forestry, or municipal waste [70].

¹⁷ The second generation scenario (2GEN) was not reassessed taking climate change into consideration. This is because results showed that the first generation scenario (1GEN) is economically more attractive. This would not change when considering climate change.

¹⁸ One scenario explicitly considers the introduction of a new crop instead of growing sugar cane at Medine and F.U.E.L.

¹⁹ Recall that hotels are already obliged to provide provisions for desalination plants since 2005 [37],[45].

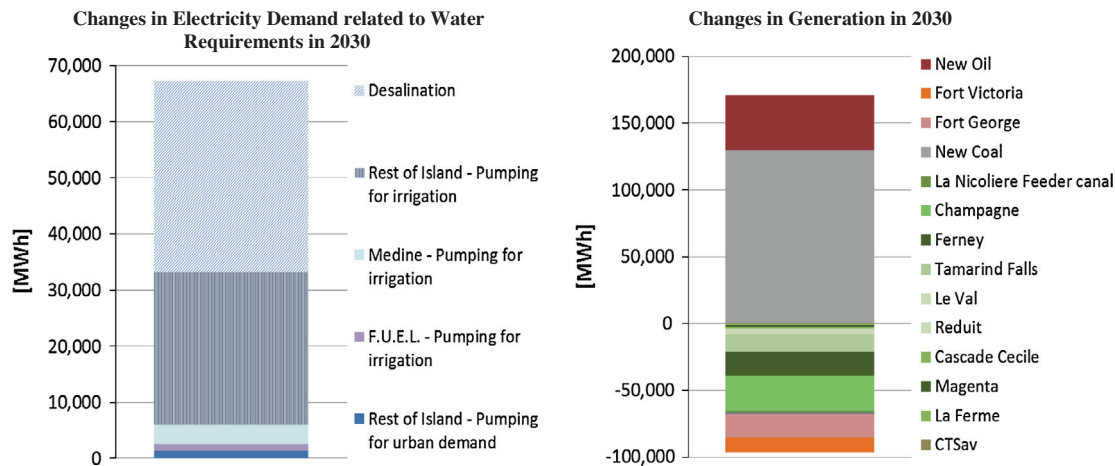


Fig. 4. Changes in electricity demand and generation for BAU + CC^{CLEWS} compared to BAU^{CLEWS}.

potential conclusions vary when considering the more integrated CLEWS approach.

5.1. Business as usual

5.1.1. Current Practice approach

Based on the calibration with historical values and future expansion plans, Fig. 3 shows the generation mix over the modelling period (BAU). Electricity generation increases from 2240 GWh in 2005 to 5260 GWh in 2030. Most future demand is met by coal-fired power plants (grey areas), with a significant reduction of the share of oil-fired power generation (red areas). This shift from petroleum products to coal is in line with the government's strategy to diversify its electricity mix and to avoid oil. Reducing the dependence on oil serves to minimise the associated higher geopolitical risks and more volatile prices.

The BAU scenario was reassessed taking the effects of climate change into account (BAU + CC). Its impact was found to be insignificant when looking at overall generation levels. However, this changes when focusing specifically on hydropower: reductions of 40% from 97,500 MWh to 58,900 MWh occur in 2030 (BAU vs. BAU + CC). These reductions are mainly compensated through increases in coal, followed by oil-fired power plants. This results in an increase of greenhouse gas emissions by 42,000 tons of CO₂-eq in 2030 compared to BAU. This increase constitutes just over one per cent of Mauritius' total net emissions [43].

5.1.2. CLEWS approach

The dynamics of the BAU scenario without climate change remain very similar once reassessed with the holistic CLEWS

approach (BAU^{CLEWS}). There are only some minor changes in hydropower generation towards the end of the modelling period due to increases in urban consumption. However, when reassessing the BAU taking climate change into consideration (BAU + CC^{CLEWS}), several dynamics would have been overlooked: through the CLEWS approach an additional electricity demand for water supply is identified, which amounts to 67,000 MWh in 2030. This is due to sea water desalination and pumping to meet urban water demand and as well groundwater pumping for irrigation (left graph of Fig. 4).

Further, due to the consideration of climate change, the hydropower generation is reduced from 83,200 MWh to 16,200 MWh in 2030 (BAU^{CLEWS} vs. BAU + CC^{CLEWS}). This 81% reduction is much more significant than the 40% reduction as observed in the corresponding scenarios of the Current Practice approach (BAU vs. BAU + CC). The Current Practice approach would therefore lead to an over 260% higher hydropower generation in 2030. This may not considerably affect the total electricity generation given the limited role of hydropower for Mauritius. However, for countries with higher hydropower shares such dynamics should not be overlooked.

Fig. 5 depicts the reasons for this overestimation. As identified by the water model, withdrawals from reservoirs for urban and agricultural water demand increase when considering climate change. This accelerates the drawdown of the reservoirs, leaving little water to be used for hydropower generation (graph on the left of Fig. 5). Without the CLEWS approach, these increasing withdrawals were overlooked and all reservoirs are able to recover once they have been emptied in summer, as shown on the right of Fig. 5.

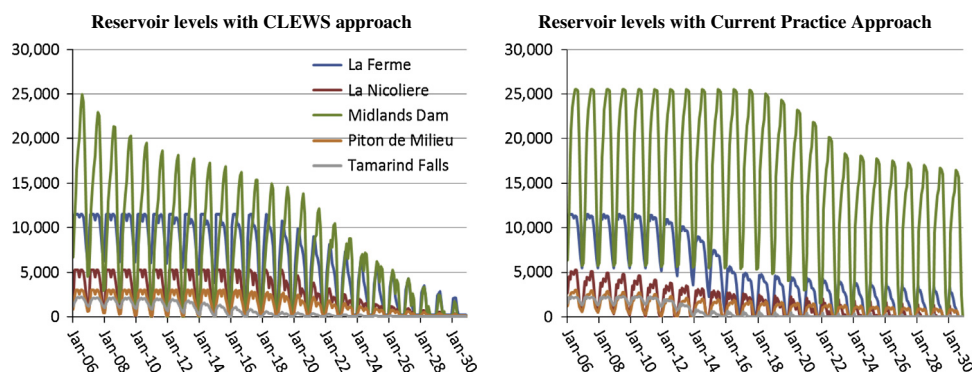


Fig. 5. Projected reservoir levels in thousand m³.

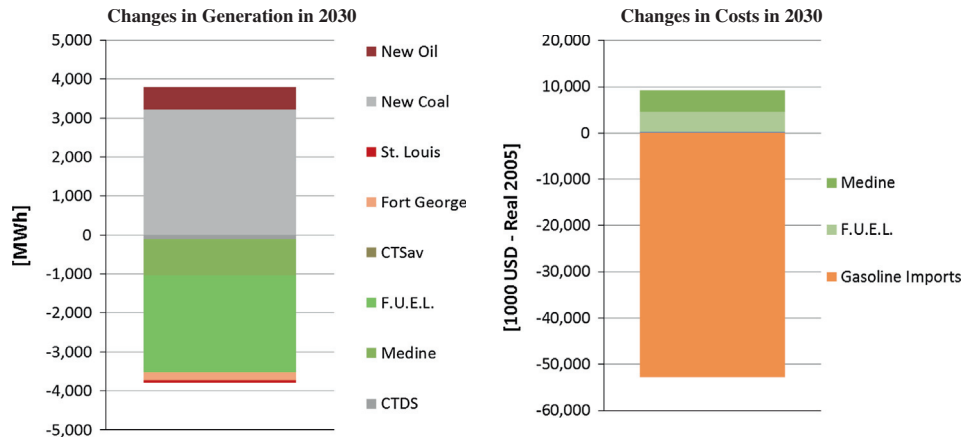


Fig. 6. Changes in selected energy dynamics for 1GEN compared to BAU.

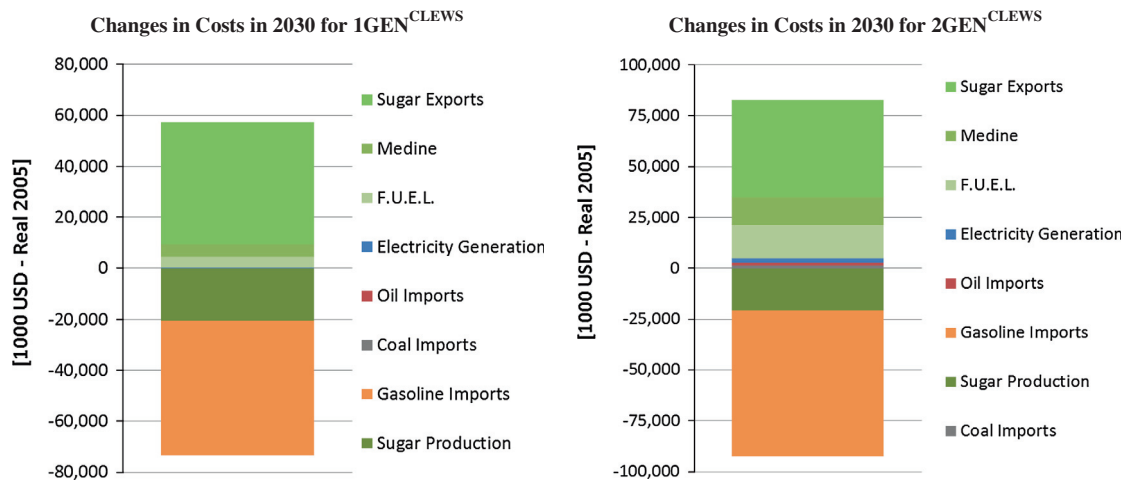


Fig. 7. Changes in costs for 1GEN^{CLEWS} and 2GEN^{CLEWS}, both compared to BAU^{CLEWS}.

The additional demand and lower hydropower generation when considering climate change result in changes in generation as illustrated for 2030 in the graph on the right of Fig. 4. The bars with negative y-axis values show power plants which decrease their generation in the BAU + CC^{CLEWS} scenario compared to the corresponding business as usual scenario (BAU^{CLEWS}). Similarly, the bars with positive y-axis values represent power plants which are expected to increase their output: these are oil and coal-fired power plants, which leads to an increase in greenhouse gas emissions of 154,000 tons of CO_{2-eq} compared to the corresponding business as usual case without climate change.

5.2. Ethanol – 1st generation

5.2.1. Current Practice approach

In this scenario it was assumed that the sugar cane processing plants Medine and F.U.E.L. were converted to produce first generation ethanol from sugar cane (1GEN). The ethanol would be used to replace some of the gasoline required for the car fleet. 1950 TJ of ethanol per year could be produced by the two plants. This could be used to reduce gasoline imports or for export. It compares to a total demand for gasoline of 5350 TJ in 2010 [30]. Due to the higher electricity requirements for producing ethanol as compared to sugar, the two sugar cane processing plants would export 3400 MWh less electricity to the national grid. This would be

compensated by increased generation from coal and oil-fired power plants, as illustrated for 2030 in the left graph of Fig. 6.

However, the resulting greenhouse gas balance is very much in favour of the ethanol production. 148,000 tons of CO_{2-eq} could be avoided in 2030. This is due to the use of ethanol as a substitute for gasoline in the transport sector and the reduced external emissions of associated oil refining. Also from an economic perspective, when focusing solely on the energy sector, this measure would be very favourable. The reduced gasoline imports outweigh the cost increases for the ethanol production at Medine and F.U.E.L. by far, as shown on the right of Fig. 6. The total net benefit of 43.5 million USD in 2030 is derived by subtracting the benefit due to the reduced imports from the additional costs at Medine and F.U.E.L.

Next, the first generation scenario is reassessed taking climate change into consideration (1GEN + CC). The corresponding changes in generation are due to reductions in hydropower (as observed in BAU + CC). This is compensated by additional generation from coal and oil-fired power plants. The ethanol production of 1950 TJ remains the same as in the 1GEN scenario. Greenhouse gas reductions of 106,000 tons of CO_{2-eq} occur in 2030. Given the lower hydropower generation, these reductions are less significant than in the case without climate change (1GEN). From an economic perspective, the dynamics do not change significantly due to climate change (i.e., compared with the right graph of Fig. 6). The additional costs for coal and oil imports to compensate the reduced

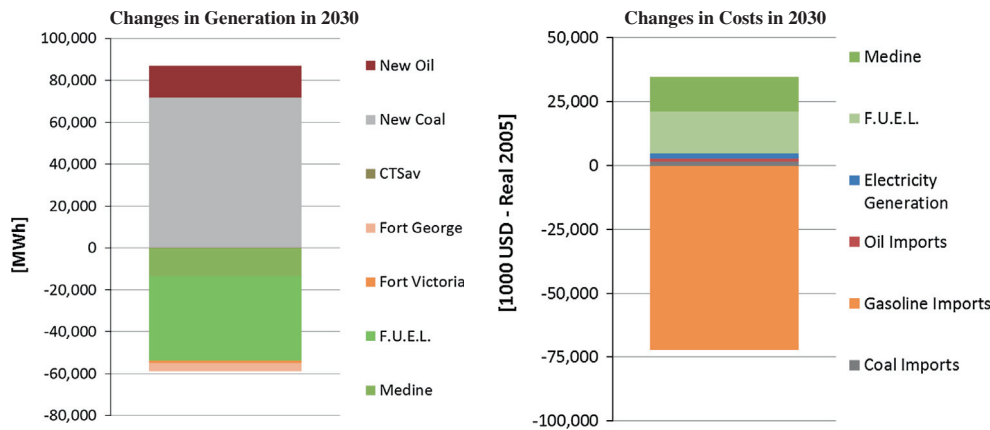


Fig. 8. Changes in selected energy dynamics for 2GEN compared to BAU.

hydropower generation are negligible compared to the benefits associated with the reduced gasoline imports and the additional costs at Medine and F.U.E.L. Ethanol generation therefore remains attractive.

5.2.2. CLEWS approach

Next, first generation ethanol production is reassessed with the CLEWS approach. Without taking climate change into account (**1GEN^{CLEWS}**), very little interaction with the actual land-use or water system occurs.

However, from an economic perspective, there is an important interlinkage with the agricultural sector: the use of sugar cane for ethanol production reduces the income from sugar exports by 48.0 million USD. Some of this is compensated by the reduced expenses for sugar production of 20.6 million USD (Fig. 7). Thus, the total economic benefit of ethanol production is significantly reduced by 27.4 million USD²⁰. This type of economic interlinkage to another sector might be simple in this case. However, it can easily become more complex if several feedstocks compete for various uses like electricity generation, biofuel production or food processing.

When adding climate change considerations (**1GEN + CC^{CLEWS}**), the additional water demand for producing first generation ethanol from sugar cane has an insignificant impact on the additional energy demand. The related dynamics presented in the right graph of Fig. 4 therefore remain. From an economic point of view, the overall changes in costs remain similar to those of the corresponding CLEWS scenario without climate change (**1GEN^{CLEWS}**) as shown in the left graph of Fig. 7. However, additional costs of 8.2 million USD are incurred to compensate for the reduced hydropower generation. There are no greenhouse gas emission reductions relative to the BAU^{CLEWS} any longer. Rather, an increase of 10,000 tons of CO_{2-eq} occurs. This is due to the lower hydropower generation and the higher total electricity demand, which is met by coal and oil-fired power plants.

5.3. Ethanol – 2nd generation

5.3.1. Current Practice approach

If the excess bagasse is also converted to ethanol (**2GEN**), both processing plants are no longer able to export any electricity to the grid. Instead the ethanol plants have to buy electricity from the grid, resulting in a higher total electricity demand. Therefore, as illustrated by the graph on the left of Fig. 8, the increases in generation by coal and oil-fired power plants outweigh the decreases

in generation at Medine and F.U.E.L. The overall ethanol production increases to 2660 GJ at the price of slightly higher overall costs compared to the first generation scenario (**1GEN**): While costs for gasoline imports decrease, this is outweighed by the higher costs of producing second generation ethanol (graph on the right of Fig. 8, compared with graph on the right of Fig. 6). The economic benefit of producing 2nd generation ethanol amounts to 37.2 million USD in 2030, and is therefore significantly lower than for 1st generation ethanol production. The greenhouse gas reductions of 123,000 tons of CO_{2-eq} in 2030 are as well lower than in the first generation scenario. Overall, apart from the opportunity to increase Mauritius' energy security by reducing its import dependency on gasoline (or to diversify export earnings through additional ethanol exports), the first generation scenario therefore seems favourable. When pursuing the Current Practice approach, taking climate change into consideration would only affect the hydropower generation, but not the profitability of the ethanol generation. Second generation ethanol production would therefore remain less attractive.

5.3.2. CLEWS approach

As in the **1GEN^{CLEWS}** scenario, very little interaction with the actual land-use or water system occurs in the **2GEN^{CLEWS}** scenario. However, income from sugar exports and the expenditures for sugar production are reduced (left graph of Fig. 7). Thus, as in **1GEN^{CLEWS}**, the total economic benefit of ethanol production is significantly reduced by 27.4 million USD when pursuing the CLEWS approach.

When considering climate change, applying the CLEWS approach does not change the fact that 2nd generation ethanol production would remain less attractive than 1st generation ethanol production. The CLEWS approach was therefore applied in order to investigate the profitability of growing an alternative crop²¹ as ethanol feedstock for the processing plants Medine and F.U.E.L. (**2NC + CC^{CLEWS}**). Despite its two crop cycles and even as climate change is considered, the alternative crop has a lower overall energy yield and requires less electricity for pumping and irrigation than sugar cane does without taking climate change into account. The electricity demand at the two selected processing plants is therefore lower than in the business as usual case without climate change (negative bars in graph on the left of Fig. 9). This reduced water demand also translates into a lower demand for desalination compared to the other climate change scenarios (compare to right graph of Fig. 4). As sugar cane is abandoned at Medine and F.U.E.L., no bagasse is available for electricity generation and both processing plants

²⁰ However, the benefits of increasing Mauritius' energy security by reducing its import dependency on gasoline remain.

²¹ Its characteristics were aligned with those of corn.

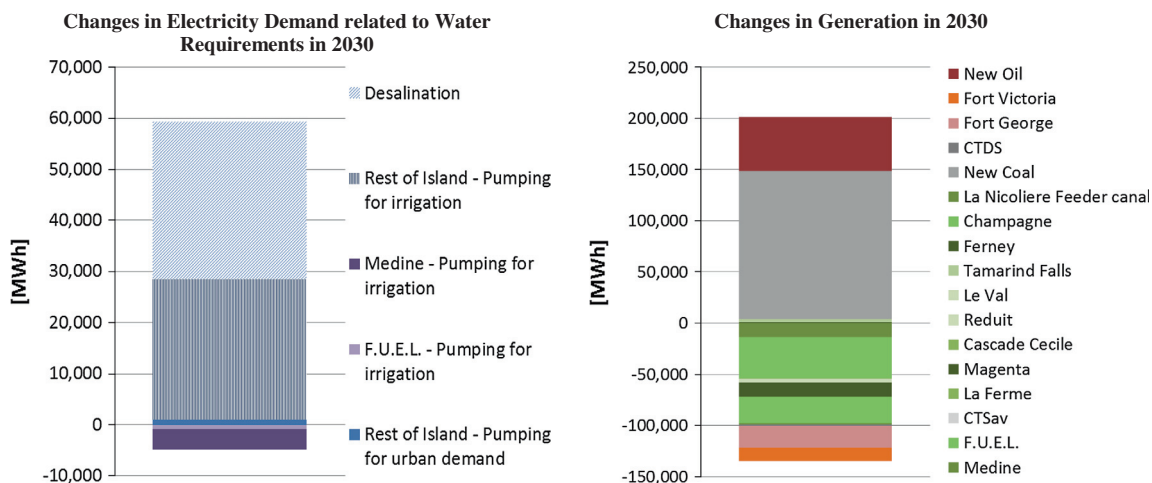


Fig. 9. Changes in electricity demand and generation for 2NC + CC^{CLEWS} compared to BAU^{CLEWS}.

need to buy electricity from the grid. This additional electricity demand is met by coal and oil-fired power generation. The overall changes in generation when introducing a new crop and considering climate change are shown in the right graph of Fig. 9.

Despite the higher energy intensity of the new crop, due to the overall lower yield only 1930 TJ of ethanol can be produced. This is almost the same yield as with first generation ethanol from sugar cane. Also the overall economic balance is less favourable than for first generation ethanol production with sugar cane. This is due to the higher processing costs associated with second generation ethanol production, the higher electricity generation requirements and the lower ethanol production. Further, greenhouse gas emissions increase to 59,000 additional tons of CO_{2-eq} as compared to the business as usual case (BAU^{CLEWS}). Overall, while diversifying agriculture is one of the priorities of the Government of Mauritius, the new crop performs worse than sugar cane from a cost and climate change perspective.

5.4. Adding value with CLEWS – summary of the findings

When assessing the energy system based on the Current Practice approach, we come to the conclusion that first generation ethanol production is very advantageous for Mauritius. It is characterized by a positive economic balance of 43.5 million USD, increased energy security through the production of 1950 TJ ethanol and 148,000 tons of CO_{2-eq} reductions in 2030. Second generation ethanol production appears less attractive. Apart from the increases in ethanol production, both the cost and greenhouse gas balances look worse.

The conclusions regarding first generation ethanol production do not change significantly when considering climate change. While hydropower generation decreases by 40%, this still constitutes an insignificant fraction of the overall electricity generation. The energy security situation compared to the business as usual deteriorates slightly. Additional coal and oil net imports total 400 TJ. The overall greenhouse gas emission reductions now amount to 106,000 tons of CO_{2-eq}.

When pursuing an integrated CLEWS approach, the picture changes considerably. The economic balance for both first and second generation ethanol production decreases by around 27.5 million USD. This corresponds to over 60% of the total net benefit for first generation and over 70% for second generation ethanol production. The reductions are mainly due to the losses in sugar exports. Greenhouse gas reductions and ethanol production remain basically the same as in the Current Practice assessment.

When assessing the climate change scenarios, the added value of the CLEWS approach becomes even more evident. These scenarios are characterized by an additional electricity demand for water supply of 67,000 MWh. The hydropower generation decreases more significantly and, in 2030, constitutes only 28% of the generation derived from the Current Practice assessment. Further, compared to the Current Practice approach, the additional net imports of oil and coal increase 3.5-fold to 1380 TJ in the first generation ethanol scenario²². This ultimately results in an increase of greenhouse gas emissions by 10,000 tons of CO_{2-eq} instead of the previous decrease. Finally, there is a whole new scenario which could not have been assessed without the CLEWS approach. This scenario focuses on the introduction of a new crop with its characteristic fertilizer, water and consequent energy demands as well as greenhouse gas emissions²³.

6. Conclusions

Judging from this and related analysis [17], a basic level of collaboration with professionals from outside of the energy sector appears valuable when setting up an energy assessment. Such collaboration helps to indicate upfront if any future resource related stresses might occur. In some cases it will become apparent that a deep integration among CLEWS will not be required. In other instances its added value might justify the modelling and coordination effort involved (refer to footnote 12).

In this study issues such as a policy to diversify agriculture, ethanol production targets, and the potential need for desalination suggested upfront that interdependencies between resource systems may require consideration in the energy model. The subsequent analysis showed that moving from water surplus to water stress increased the level of CLEWS interactions to a point where national dynamics around an ethanol production policy changed significantly.

In general, a CLEWS approach is likely to be valuable for countries intending to implement integrated policies with potential implications for multiple resource systems. High shares of hydro-

²² This refers to *additional imports* compared to the respective business as usual cases (BAU and BAU^{CLEWS}). While such a comparison is commonly used for energy models, a comparison of the total values instead of the additions would obviously yield much lower differences in per cent.

²³ The crop in this scenario did not perform better than sugar cane. However, it requires a CLEWS assessment to come to this conclusion and to justify the assumption that sugar cane is also preferable once climate change considerations are taken into account.

power and expected climate induced rainfall changes might also indicate the relevance of a CLEWS assessment. In addition to climate change, conflicting water management priorities may impede the future availability of water for electricity generation. Applying a CLEWS approach may further enable a more holistic assessment of greenhouse gas emissions. This may serve, for example, to estimate the climate implications of technological change, new agricultural practices, and ensuring energy and water supply.

The additional undertaking of interlinking the energy model with external climate, land-use and water models involves a non-trivial effort. However, the purpose of an energy model should be to depict a potential future as adequately as possible in order to help derive strategies that are consistent across sectors. The underlying research work has shown the importance of capturing interlinkages between CLEWS for informing such consistent strategies. In response to this work, the Government of Mauritius has announced the appointment of a high-level CLEWS panel. This panel will ensure an integrated approach to CLEWS-related policies [71].

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.apenergy.2013.08.083>.

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